The combustion of fossil fuel adds $CO_2$, which is almost free of $^{14}C$, to the atmosphere, thus reducing the specific activity (the dilution was about 3% for the period 1900–1970). Taking all anthropogenic sources into account, a global average specific activity of modern carbon is now $13.56 \pm 0.07$ dpm/g C. In §13.8.1 we discuss dating by the "C14-method", for which such a figure is important.

## 13.5. PRIMORDIAL RADIONUCLIDES

Some radionuclides have halflives so long that some of what was formed in the aftermath of the Big Bang and later supernova explosions still remains on the Earth, and some of their daughters.

### 13.5.1. Very long-lived nuclides lighter than lead[2]

As the detection technique for radioactivity has been refined, a number of long-lived radionuclides have been discovered in nature. The lightest have been mentioned in §13.4.1. The heavier ones, not belonging to the natural radioactive decay series of uranium and thorium, are listed in Table 13.3. $^{50}V$ is the nuclide of lowest elemental specific activity (~0.0001 Bq/g) while the highest are $^{87}Rb$ and $^{187}Re$ (each ~900 Bq/g). As our ability to make reliable measurements of low activities increases, the number of elements between potassium and lead with radioactive isotopes in nature can be expected to increase.

**Table 13.3** Primordial radionuclides for $Z < 82$ (Pb)

| Nuclide | Isotopic abundance % | Decay mode and particle energy MeV | Half-life (years) |
|---|---|---|---|
| $^{40}K$ | 0.0117 | $\beta^-$ EC 1.31 | $1.26 \times 10^9$ |
| $^{50}V$ | 0.250 | $\beta^-$ EC (0.601) | $>1.4 \times 10^{17}$ |
| $^{87}Rb$ | 27.83 | $\beta^-$ 0.273 | $4.88 \times 10^{10}$ |
| $^{115}In$ | 95.72 | $\beta^-$ 1.0 | $4.4 \times 10^{14}$ |
| $^{123}Te$ | 0.905 | $\beta^-$ EC (0.052) | $1.3 \times 10^{13}$ |
| $^{138}La$ | 0.092 | $\beta^-$ EC | $1.06 \times 10^{11}$ |
| $^{144}Nd$ | 23.80 | $\alpha$ | $2.1 \times 10^{15}$ |
| $^{147}Sm$ | 15.0 | $\beta^-$ 2.23 | $1.06 \times 10^{11}$ |
| $^{148}Sm$ | 11.3 | $\beta^-$ 1.96 | $7 \times 10^{15}$ |
| $^{176}Lu$ | 2.59 | $\beta^-$(1.188) | $3.8 \times 10^{10}$ |
| $^{174}Hf$ | 0.162 | $\alpha$ | $2 \times 10^{15}$ |
| $^{187}Re$ | 62.60 | $\beta^-$ 0.0025 | $4.2 \times 10^{10}$ |
| $^{190}Pt$ | 0.012 | $\alpha$ | $6.5 \times 10^{11}$ |

Values within parenthesis refer to decay energies.

[2] The heavier elements are discussed in §13.5.2 and in Ch. 14.

[3] The student can easily trace the decay series in Fig. 13.4 with a transparent marker.

Because of the long half-lives of these nuclides they must have been formed at the time of (or possibly even before) the formation of the solar system and of the earth. When the earth's crust solidified, these radionuclides became trapped in rocks. As they decayed, decay products accumulated in the closed rock environment. By measuring the amount of parent and daughter nuclides, it is possible with the half-life to calculate how long this environment (e.g. a rock formation) has existed. This is the bases for *nuclear dating* (also called "radioactive clocks"), and almost all of the nuclides in Tables 13.1 and 13.2 can be used for this purpose. In §13.8 we discuss dating methods for the K—Ar, Rb—Sr, and Re—Os systems.

A careful look at these naturally occurring long-lived nuclei reveals that some of them appear in *short* decay series, e.g. $^{152}$Gd → $^{148}$Sm → $^{144}$Nd → $^{140}$Ce and $^{190}$Pt → $^{186}$Os → $^{182}$W. The heavy element series beginning with U and Th isotopes are therefore referred to as *long* decay series.

### 13.5.2. Elements in the natural radioactive decay series

In Chapter 1 we briefly discussed the existence of four long series of genetically related radioactive nuclides beginning with Th, U or Np and ending with Pb or Bi. In Figure 13.4 we present all known isotopes of elements $_{81}$Tl to $_{92}$U. The chemistry and other data for Th and U are given in detail in Ch 14. Some of the nuclides occur naturally in the long decay series shown in Figure 1.2. Others are produced through nuclear reactions according to the schemes in Figure 5.7 and principles described in Ch. 10 & 17. The first series in Figure 1.2 is known as the *thorium decay series*, and consists of a group of radionuclides related through decay in which all the mass numbers are evenly divisible by four (the 4*n* series). It has its natural origin in $^{232}$Th which occurs with 100% isotopic abundance. Natural thorium has a specific activity *S* of 4.06 MBq/kg, as its half-life through α-decay is $1.41 \times 10^{10}$ y. The terminal nuclide in this decay series is the stable species $^{208}$Pb (also known as ThD). The transformation from the original parent to the final product requires 6 α- and 4 β-decays. The longest-lived intermediate is 5.76 y $^{228}$Ra.

The *uranium decay series* consist of a group of nuclides that, when their mass number is divided by 4, have a remainder of 2 (the $4n + 2$ series). The parent of this series is $^{238}$U with a natural abundance of 99.3%; it undergoes α-decay with a half-life of $4.46 \times 10^9$ y. The stable end product of the uranium series is $^{206}$Pb, which is reached after 8 α- and 6 β-decay steps.

The specific activity of $^{238}$U is 12.44 MBq/kg $^{238}$U. However, because *natural uranium* consists of 3 isotopes, $^{238}$U, $^{235}$U and $^{234}$U, whose isotopic abundances are 99.2745%, 0.7200% and 0.0055%, respectively, the specific activity of natural uranium is 25.4 MBq/kg.

The uranium decay series provides the most important isotopes of elements radium, radon, and polonium, which can be isolated in the processing of uranium minerals. Each ton of uranium is associated with 0.340 g of $^{226}$Ra. Freshly isolated $^{226}$Ra reaches

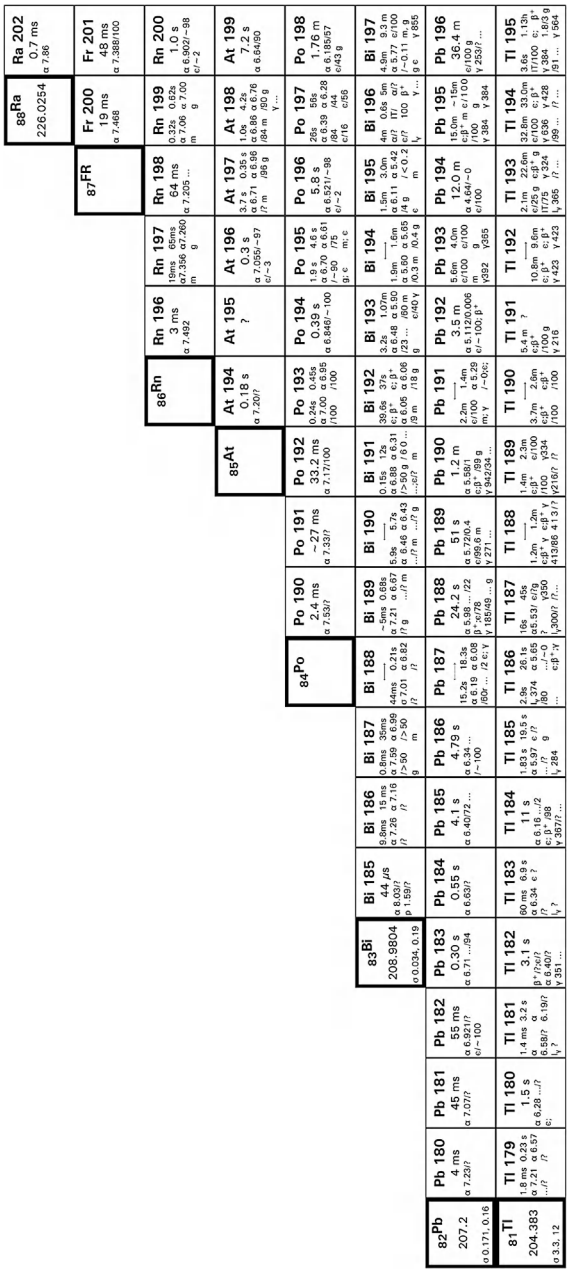



**Figure 13.4** Chart of all known isotopes of elements $_{81}$Tl to $_{92}$U. The legend to the chart is given in Fig. 5.7.

**Figure 13.4** (*continued*).

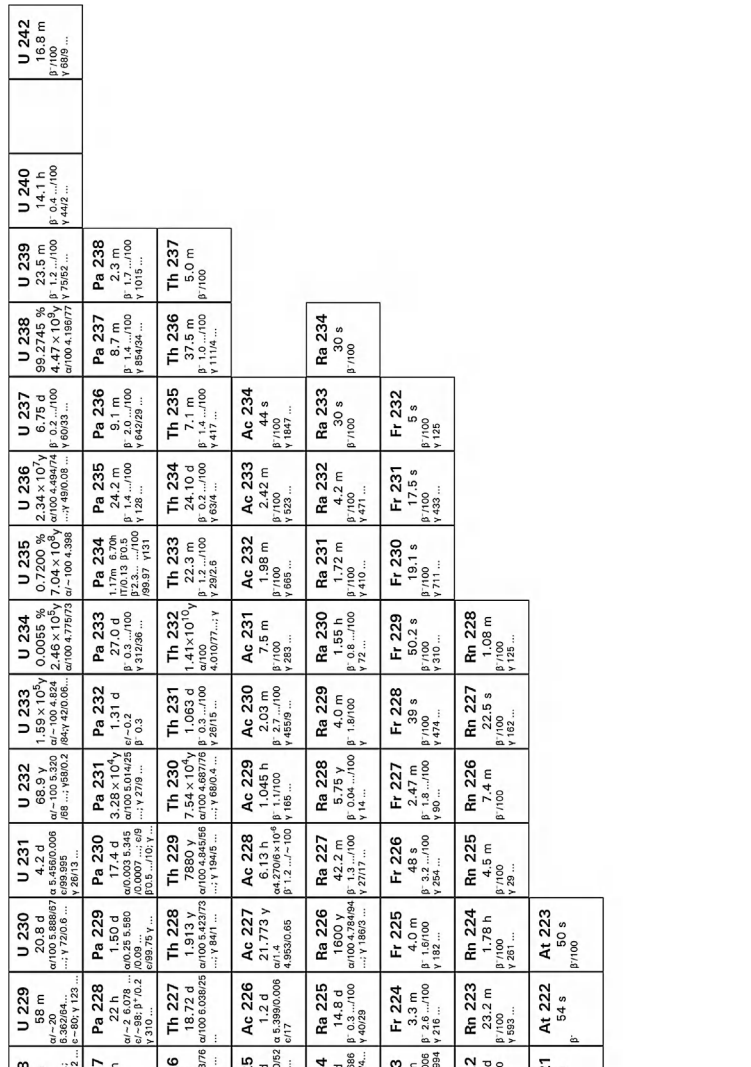



**Figure 13.4** (*continued*).

radioactive equilibrium with its decay products to $^{210}$Pb in about two weeks (see Fig. 1.2). Many of these products emit energetic γ-rays, which resulted in the use of Ra as a γ-source in medical treatment of cancer (radiation therapy). However, the medical importance of radium has diminished greatly since the introduction of other radiation sources, and presently the largest use of radium is as small neutron sources (see Table 10.2).

Although the chemistry of radium is relatively simple (like barium), the fact that it produces a radioactive gas (radon) complicates its handling. The decay of radon produces "airborne" radioactive atoms of At, Po, Bi, and Pb. Since uranium is a common element in rocks it is also a common constituent of building materials. Such material emits Rn, as discussed further in §13.6. Work with radium compounds should be carried out within enclosures to avoid exposure to Rn and its daughters. In recent years the use of $^{223}$Ra has become increasingly common when treating bone cancer. The use is both for easing the pain but also for direct treatment. Radium is an alkali metal with the homologues Ca, Sr and Ba and substitutes readily with these, e.g. with the Ca in bone.

The *actinium decay series* consists of a group of nuclides whose mass number divided by 4 leaves a remainder of 3 (the $4n + 3$ series). This series begins with the uranium isotope $^{235}$U, which has a half-life of $7.04 \times 10^8$ y and a specific activity of $8 \times 10^4$ MBq/kg. The stable end product of the series is $^{207}$Pb, which is formed after 7 α- and 4 β-decays. The actinium series includes the most important isotopes of the elements protactinium, actinium, francium, and astatine. Inasmuch as $^{235}$U is a component of natural uranium, these elements can be isolated in the processing of uranium minerals. The longest-lived protactinium isotope, $^{231}$Pa ($t_{½}$ $3.28 \times 10^4$ y) has been isolated on the 100 g scale, and is the main isotope for the study of protactinium chemistry. $^{227}$Ac ($t_{½}$ 21.8 y) is the longest-lived actinium isotope.

A fourth long radioactive decay series, the *neptunium series* (Fig. 1.2), is composed of nuclides having mass numbers which divided by 4 have a remainder of 1 (the $4n + 1$ series). The name comes from the longest lived $A = 4n + 1$ nuclide heavier than Bi, $^{237}$Np, which is considered as the parent species; it has a half-life of $2.14 \times 10^6$ y. In as much as this half-life is considerably shorter than the age of the earth, primordial $^{237}$Np no longer exists on earth, and, therefore, the neptunium series is not found as a natural occurrence. Np found in nature on Earth has now human origin. However, Np has been discovered in the spectrum of some stars. All known Np-isotopes are presented in Figure 14.1.

Very small amounts of $^{237}$Np, as well as of $^{239}$Pu, have been discovered on earth; the half-lives of $^{239}$Pu (in the $4n + 3$ series) is $2.411 \times 10^4$ y. Both isotopes are too short-lived to have survived the 4 eons since the solar system was formed. However, they are always found in minerals containing uranium and thorium and it is believed that the neutrons produced in these minerals through (α,n) and (γ,n) reactions with U and Th as well as by spontaneous fission of $^{238}$U form the neptunium and plutonium through n-capture and β-decay processes. The n-production rate in the uranium mineral pitchblende (containing ~50% U) is about 50 n/kg s. The typical value for the $^{239}$Pu/$^{238}$U ratio in minerals is $3 \times 10^{-12}$.

The end product of the neptunium series is $^{209}$Bi, which is the only stable isotope of bismuth. Seven α- and four β-decays are required in the sequence from the parent $^{237}$Np to $^{209}$Bi. An important nuclide in the neptunium decay series is the uranium isotope $^{233}$U, which has a half-life of $1.59 \times 10^5$ y (the most stable intermediate) and, like $^{235}$U, is fissionable by slow neutrons.

The long-lived plutonium isotope $^{244}$Pu (belonging to the $4n$ series; see also Fig. 14.1), which decays through α-emission and spontaneous fission (0.13%) with a total half-life of $8.26 \times 10^7$ y, was discovered in rare earth minerals in 1971. If this is a survival of primeval $^{244}$Pu, only $10^{-15}$ % of the original can remain. An alternate possibility is that this $^{244}$Pu is a contaminant from cosmic dust (e.g. from a supernova explosion in more recent times than the age of the solar system).

## 13.6. RADIUM AND RADON IN THE ENVIRONMENT

In uranium ore radioactive equilibrium is established between the mother $^{238}$U and daughters in the decay chain (see Fig. 1.2, $4n + 2$ series) at *the rate of the daughter half-life* and at the *level of the mother decay rate*. The chain passes $^{226}$Ra and $^{222}$Rn and daughters down to $^{206}$Pb at a rate corresponding to the original amount of $^{238}$U, as long as the host material is undisturbed. The equilibrium between $^{226}$Ra and $^{210}$Pb is established within some weeks. Since $^{222}$Rn is a gaseous intermediate, its daughters are likely to be formed in air ("radon daughters"). Radon diffuses out of thorium and uranium minerals, and adds radioactivity to the ground water and to the atmosphere both by its own presence and that of its daughters. Since Ra and Rn are among the most radio-toxic substances existing, causing bone and lung cancer at relatively low concentrations (the risk levels are discussed in Ch. 15), special attention must be devoted to their appearance in nature.

Common Rn-concentrations in ground water are 5–300 kBq/m$^3$, but in areas of U-rich granite values $\gg$1 MBq $^{222}$Rn/m$^3$ occur. Tap water usually contains of the order of 1 kBq/m$^3$. In many places water from hot mineral wells is considered beneficial to health both for bathing and for drinking ("spas" or hot springs). The water may be warm due to radiogenic heating at the source (minerals rich in U or Th) and have a high content of dissolved radium and radon. Thus, in famous spas in Europe the $^{222}$Rn concentration for "therapeutic inhalation" may be 1 MBq/m$^3$ air (Baden-Baden). From the Joachimsthal U-mine in Bohemia, which contains a number of hot wells (29°C), water containing 10–15 MBq $^{222}$Rn/m$^3$ is pumped to spas, where it is used as medical treatment against rheumatism (a 30 min bath a day).

The average exhalation rate of radon from the ground is 5–50 mBq/m$^2$s, leading to a near ground level radon concentration of 1–10 Bq/m$^3$, but varies widely with ground conditions. In soil over Swedish uranium bearing shale ($\approx$300 ppm U) the $^{222}$Rn concentration can exceed 1 MBq/m$^3$, though on the average the concentration in Scandinavian air (the main geology consists of granitic rocks) is only 3 Bq/m$^3$.